Highlights

RansKnow: An Open Dataset of Ransomware-Related YouTube Transcripts with Regex-Extracted MITRE ATT&CK Features

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- Open corpus of 1,128 ransomware-related YouTube video records from 92 channels
- 9 of 41 ransomware family aliases collide with ordinary English or business words
- Weak-label baselines reach macro-F1 0.90 on relevance and 0.38 on dominant tactic
- Softmax OOD detectors fail a sanity check that embedding distance passes at 0.94
- Corpus, extraction pipeline, audit and experiment code are released openly

RansKnow: An Open Dataset of Ransomware-Related YouTube Transcripts with Regex-Extracted MITRE ATT&CK Features

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Abstract

Security professionals share much of their ransomware knowledge in conference talks and vendor webinars, but we are not aware of an open transcript-level corpus that covers this material. We present RansKnow, an open corpus of 1,128 ransomware-related YouTube video records (1,034 with transcripts) from 92 cybersecurity channels, with MITRE ATT&CK-aligned family, tactic, tool, and platform features extracted by regular expressions. A label-quality audit found that 9 of 41 family aliases collide with ordinary words such as `play` and `hive`, which inflated the most-detected family about 3.5× before a context-based fix. The corpus has no independent human-annotated test set, so baseline, calibration, and conformal results measure agreement with the audited pipeline, not real-world accuracy. On that basis, baselines reach macro-F1 0.90 on relevance, 0.62 on platform, 0.38 on dominant tactic, and at most 0.20 on family and tool. Split conformal prediction reaches 0.91–0.93 coverage at a 90% target on the multiclass tasks. In out-of-distribution detection, softmax methods fail a sanity check against off-topic newsgroup posts (AUROC 0.62), while Mahalanobis distance on embeddings passes it (0.94 after controlling for length and formatting) but scores below chance on two setups whose out-of-distribution videos are selected for naming a family. We release the corpus, pipeline, and experiment code. The audit method applies to other security text datasets labeled by keywords or regular expressions.

Keywords: Ransomware, Cyber threat intelligence, MITRE ATT&CK, Weak supervision, Label-quality audit, Conformal prediction, Out-of-distribution detection

1. Introduction

Ransomware knowledge is scattered across incident-response writeups, conference talks, and vendor threat-intelligence reports, and little of it is structured for NLP or machine learning. Structured resources exist, including the MITRE ATT&CK framework [1] and datasets built from written threat intelligence (Section 2), but we are not aware of an open transcript-level corpus covering what security professionals say in talks, webinars, and walkthroughs.

YouTube gathers this material from incident-response vendors, security conferences, and independent researchers, whose incentives and house styles differ in a way no single vendor blog or threat feed replicates. Transcription is the first obstacle: 92.7% of this corpus needed

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automatic speech recognition rather than official captions (Section 3.3), and a single misheard term can change a jargon-based label (Section 4.2).

Labeling is the second. RansKnow's labels come from a regex pipeline, and the project has no human-annotated evaluation set held out from that pipeline (Section 4.3). Results from Section 5 through Section 7 therefore measure agreement with the audited pipeline, not verified accuracy.

This paper makes five contributions:

1. **A corpus.** RansKnow contains 1,128 video records, 1,034 with transcripts, from 92 cybersecurity YouTube channels, with MITRE ATT&CK-aligned features (Section 3). The corpus and code are at `https://github.com/Henrinnes/RansKnow`.
2. **A label-quality audit.** A dictionary-driven search for generic-word alias collisions found that 9 of 41 family aliases collide with ordinary words, which inflated the most-detected family about 3.5× before a disambiguation fix (Section 4, Table 1).
3. **Weak-label baselines.** We report results across five prediction tasks, three feature representations, and five models, with calibration and conformal prediction on top (Sections 5 to 7).
4. **Out-of-distribution results.** These scores need no correct dominant-tactic label, although two of the four setups define their out-of-distribution populations with the regex (Section 8).
5. **Counterfactual and perturbation checks.** A classifier trained on the fixed labels and scored on videos it did not train on still ranks the videos the fix removed slightly above other negatives (AUROC 0.58), a weak trace of the fragility the audit found (Section 9).

2. Related Work

Ransomware research. Recent surveys concentrate on ransomware behavior and defense: Beaman et al. [2] review advances in prevention and detection, and Oz et al. [3] survey ransomware evolution, taxonomy, and defense solutions. Meland et al. [4] study the ransomware-as-a-service economy through two years of observation of darknet markets and forums. RansKnow addresses a different question: how practitioners describe ransomware in public talks and walkthroughs, and how reliably that description can be turned into structured data.

Mapping threat reports to ATT&CK. A related line of work extracts structured behavior from written reports. TTPDrill [5] extracts threat actions from threat reports and aligns them with attack patterns, rcATT [6] compares classifiers for retrieving ATT&CK tactics and techniques from reports and releases a tool based on the results, and Orbinato et al. [7] report an experimental study of machine-learning classification of unstructured threat intelligence into attack techniques. These works take written text as input. RansKnow applies a deterministic regex method to mostly ASR transcripts and audits the labels that method produces.

Ransomware and cyber-threat-intelligence datasets. RansKnow's tactic taxonomy follows the MITRE ATT&CK framework [1], and prior datasets build on that framework using written cyber-threat-intelligence (CTI) text. MalwareTextDB [8] provides span-level

annotations of malware characteristics (39 APT reports, 6,819 sentences), and AnnoCTR [9] links entities, tactics, and techniques across 400 commercial CTI reports. Both rely on human annotation of written vendor reports and blog posts. RansKnow differs in two ways. Its source material is spoken narration, 92.7% of it transcribed by ASR (Section 3.3), from vendor analysts, conference presenters, and independent researchers. Its features come from an audited regex and keyword pipeline (Section 4) rather than human annotation, which gives weaker labels in exchange for scale and openness. We are not aware of a comparable open, transcript-level, MITRE-aligned video corpus, although our search was not a systematic review.

Weak and distant supervision. Replacing human labels with an automated heuristic is a well-studied tradeoff. Mintz et al. [10] formalized distant supervision for relation extraction by aligning text against a knowledge base, accepting label noise in exchange for scale, and Ratner et al. [11]'s Snorkel combines multiple, possibly conflicting, weak labeling functions probabilistically. The Knowledge Agent (Section 4) is a single deterministic labeling function. That makes it simpler to audit: the alias-collision analysis in Section 4.1 would be harder against a black-box combination of weak labelers. It also rules out cancelling noise through redundancy, which motivates the scope note in Section 4.3. Label-quality research reaches similar concerns from the model side: confident learning [12] characterizes and identifies label errors in datasets, and Northcutt et al. [13] estimate an average of 3.3% label errors across the test sets of 10 widely used benchmarks and show that such errors can destabilize benchmark results. RansKnow's audit starts from the labeling function instead, searching its alias list for collisions with ordinary vocabulary.

Conformal prediction and out-of-distribution detection. Split conformal prediction [14, 15] offers a distribution-free coverage guarantee (Section 7) and has seen growing use in NLP and security as an alternative to ad hoc confidence thresholds. Angelopoulos and Bates [16] survey the methods that Section 7.1 draws on. Out-of-distribution detection (Section 8) with softmax-confidence baselines [17], energy scores [18], and class-conditional embedding distance [19] is less explored for weak-label security NLP classifiers. Section 8.2 shows that confidence-based methods can fail there in a domain-specific way: a classifier's most confident predictions coincide with the "nothing detected" signature that off-topic input also produces.

3. Dataset Construction

RansKnow [20, 21] consists of 1,128 video records from 92 cybersecurity YouTube channels. Of these, 1,034 have an extracted transcript with real content (four videos are listed twice, under two registry entries for the same organization, so the 1,034 transcripts cover 1,030 distinct videos). The remaining 94 have no usable transcript.[1]

3.1. Channel selection

We curated channels in two batches. Batch 1 (50 channels, `C01`–`C50`) covers established DFIR vendors, security conferences, and independent researchers. Batch 2 (51 candidate

[1]We checked this against the released `transcripts/` tree: no method in the fetch cascade (Section 3.3) produced usable text for any of the 94, and none was excluded after fetching for another reason.

channels, C51–C101) expanded coverage into threat-intelligence vendors, additional conferences, and OT/ICS-security and government channels. Of the 51 candidates, 42 yielded usable transcripts, 1 was a confirmed duplicate of a Batch 1 channel, 4 had no discoverable dedicated YouTube channel, and 4 resolved to real channels with no ransomware-keyword-matching content in their upload history. Channel-level provenance is released with the corpus in `rubrics/Dataset_Channel_Registry_Updated_50_fixed_urls.xlsx` (Batch 1) and `rubrics/Dataset_Channel_Registry_Batch2.xlsx` (Batch 2).

3.2. Video selection

Within each channel, we scanned recent uploads for a fixed 21-term keyword list (`ransomware`, `extortion`, `lockbit`, `incident response`, ...; full list in Appendix A) matched against title and description, with a cap of 20 videos per channel and a minimum duration of 5 minutes to exclude short clips and trailers. This is a convenience sample, not a random or exhaustive one. A video is in scope only if its *title or description* contains one of the 21 terms, which conditions the corpus on each channel operator's titling and description habits. A channel that discusses ransomware extensively but titles episodes generically (e.g. by client-engagement number rather than threat name) would be under-represented, and channels that name high-profile families (`lockbit`, `conti`, ...) in titles for engagement would be over-represented. Keyword-based inclusion therefore adds a second bias on top of channel curation (Section 10).

3.3. Transcript acquisition

We acquired transcripts through a three-stage cascade: (1) the YouTube Data API's official captions where available; (2) `yt-dlp`'s auto-generated captions; (3) local Whisper transcription [22] as a fallback. Only 74 of 1,034 transcripts (7.2%) came from official captions. Another 959 (92.7%) are ASR output: 554 from `whisper_base` (the smallest, least accurate Whisper checkpoint) and 405 from a local Whisper fallback. One transcript came from `yt-dlp` captions. This composition is Finding B of the label-quality audit, and Section 4.2 reports its effect on jargon-term detection.

4. Label-Quality Audit

The Knowledge Agent pipeline (`Scripts/knowledge_agent.py`) extracts structured features from every transcript with regex matching: ransomware family mentions (against a maintained alias list), MITRE ATT&CK tactic mention counts across 10 tactics, offensive tool mentions across 8 tools, and platform signals (Windows/Linux/ESXi). We report three problems here because a classifier trained on these labels inherits their bias.

4.1. Finding A: generic-word alias collisions

Nine of 41 family aliases collide with common English or business words (`play`, `hive`, `cactus`, `agenda`, `clop`, `inc`, `maze`, `medusa`, `royal`). We identified them with a dictionary lookup against `/usr/share/dict/words` and confirmed them by inspecting transcripts (e.g. `hive` matching "registry hive," a routine DFIR term, rather than Hive ransomware). A windowed-context fix, which requires a ransomware-context term within ∼15 tokens of the match, substantially reduces false-positive volume but does not eliminate it: false positives

still occur where context vocabulary is dense (see `Scripts/audit_family_aliases.py` for the method and Table 1 for before/after counts). We have not measured the precision and recall of the fixed labels against human judgment, so Table 1 shows how many detections the fix removed, not how many of the remaining detections are correct.

Table 1: Before/after family mention counts for the 9 flagged aliases, windowed-context disambiguation applied. `Play` alone accounted for 386 of 465 (83%) of the family-tagged videos in the audit before the fix.

Family	Before	After	Dropped
Play	386	111	275
Qilin	88	21	67
Hive	23	10	13
Maze	11	5	6
Royal	28	19	9
INC	10	7	3
Cactus	9	6	3
Cl0p	11	11	0
Medusa	7	7	0

4.2. Finding B: ASR-dominated transcript quality

As noted in Section 3.3, 92.7% of transcripts are ASR output rather than official captions. Aggregate detection rates across `Transcript_Provider` show no obvious gap, but mistranscribing rare jargon can matter without moving an aggregate rate: a single missed `rclone` mention changes that video's `Tool_Rclone` feature from 1 to 0.

The released dataset contains transcripts, not audio, so we automated the planned audio-versus-transcript check. For a random sample of 40 `whisper_base` videos, we re-fetched the opening 6 minutes of audio, re-transcribed it with Whisper's `small` checkpoint as a stronger reference, and checked every occurrence of six jargon terms (`LockBit`, `Mimikatz`, `PsExec`, `Cobalt Strike`, `rclone`, `BloodHound`, plus known phonetic-mangling variants) against the stored `whisper_base` transcript. Of the 40 videos, 35 were re-transcribed (5 audio downloads failed). Across those 35, the six terms appeared in only 3 (video, term) instances within the sampled opening 6 minutes, so jargon density is low even in ransomware-focused content. The `whisper_base` transcripts missed 1 of the 3 entirely: `rclone` appears once in one video's reference transcript and zero times (correct spelling or a known mangled variant) in its stored transcript. The other two, `bloodhound` (same video) and `cobalt strike` (18 mentions in another video), were detected correctly. A miss rate of 1/3 from 3 instances is not a precise estimate. The conclusion is qualitative: jargon mistranscription is an observed failure mode of this corpus's dominant transcription method, consistent with Section 9.3's simulated-ASR-noise sweep. A larger spot-check over full video length is the natural follow-up.

4.3. Scope: no independent ground truth

An earlier design of this project planned a held-out, independently annotated evaluation subset (180 videos, stratified by channel type, year, and current family count) to break the circularity of evaluating a classifier against the pipeline that labeled its training data. That

subset was never annotated. Recruiting external annotators would have required ethics approval beyond the project's existing approval for public data, and was not undertaken. We considered a single-investigator pass and rejected it as too close to the system under evaluation to serve as independent ground truth.

As a result, **every quantitative result in Section 5 onward, apart from the population-separation scores in Section 8 and the behavioral checks in Section 9, measures agreement with the audited Knowledge Agent extraction pipeline (Section 4), not verified accuracy against human judgment.** We repeat this point where each affected result is reported. Finding A shows that the pipeline's labels contain error even after the disambiguation fix, and nothing downstream corrects for it. Macro-F1 and calibration numbers in this paper describe how well models reproduce a specific, imperfect, documented extraction process. They are not claims about ransomware-domain understanding. Section 10 returns to this limitation.

5. Task Definitions and Baselines

Five prediction tasks are defined over the corpus: ransomware family (multi-label, 18 classes after excluding 3 singleton-occurrence classes, see Table 2), dominant MITRE tactic (multi-class, 11 classes), platform signal (multi-class, 4 classes), offensive tool mentions (multi-label, 7 classes after excluding `MegaNZ`, which has zero occurrences in the corpus), and a ransomware-relevance binary screen (derived, not a stored column: `Family_Count > 0 OR Tactic_Impact > 0`).

Table 2: Task taxonomy.

Task	Type	Classes	Label source
Family	multi-label	18	alias regex (§4.1)
Dominant tactic	multi-class (+none)	11	max of 10 tactic counts
Platform signal	multi-class (+none)	4	max of 3 platform counts
Tool mentions	multi-label	7	per-tool regex
Ransomware-relevant	binary	2	derived

5.1. Model ladder and validation protocol

Each task is evaluated over two feature representations: TF-IDF and mean-pooled sentence embeddings.[2] Structured Knowledge Agent columns are added for `family` and `dominant_tactic` only. We compare five models (the rule-based Knowledge Agent as baseline 0, Logistic Regression, Linear SVM, Random Forest, LightGBM [24]) under three

[2]384-dim `all-MiniLM-L6-v2`, chunked at ∼200 words per chunk and mean-pooled, computed in a dedicated Python environment (`torch` 2.5.1) because the reference implementation's default environment ships an older `torch` that does not satisfy `sentence_transformers`' [23] requirements; see `Scripts/precompute_embeddings.py`. The 20 Newsgroups embeddings and the experiments rerun for this version used Python 3.12.3 with `torch` 2.15.0, `sentence_transformers` 6.2.0, scikit-learn 1.9.1, and LightGBM 4.7.0.

cross-validation protocols: stratified random k-fold, `GroupKFold` by `Channel_ID` (to detect channel-style leakage, since a model can learn "this is CrowdStrike's narration style" instead of the content), and a temporal split (train pre-2024, test 2024+, since 79% of the corpus is from 2024 onward).

5.2. Results

Per Section 4.3, every number in this section is weak-label agreement: how well a model reproduces the Knowledge Agent's own extraction. The rule-based Knowledge Agent baseline scores a tautological macro-F1 of 1.000 on every task because it is checked against its own labels, so Table 3 omits it. Structured features are excluded from the dominant-tactic comparison for a related reason: they contain the tactic counts its label is the argmax of. That combination scores 0.813 with LightGBM, which measures how well a tree model learns an argmax over its own inputs, so Table 3 reports the best representation without those counts.

Table 3: Best stratified-CV macro-F1 per task, across the eligible representations and trained models (LogReg, Linear SVM, Random Forest, GBM). Weak-label agreement, not verified accuracy (§4.3). Mean ± std is across 5 stratified-random split seeds (0–4) at the reported (representation, model) cell, model `random_state` held fixed (see the paragraph below the table).

Task	Representation	Model	Macro-F1 (mean ± std)
Ransomware-relevant	TF-IDF	GBM	0.897 ± 0.006
Platform signal	TF-IDF	GBM	0.616 ± 0.008
Dominant tactic	TF-IDF	GBM	0.377 ± 0.009
Tool mentions	TF-IDF	LogReg	0.166 ± 0.032
Family	embedding	LogReg	0.115 ± 0.019

The single-seed point estimates used elsewhere in this paper are the seed-0 values of the same cells, within 0.015 of the 5-seed mean for every task except `tool`, whose seed-0 value (0.200) sits 0.034 above its mean. Std is smallest for `relevance` (0.006) and largest for `tool` (0.032, ∼20% of its mean), consistent with `tool`'s instability across CV protocols (Table 4). Only the stratified-random split varies with the seed. `channel_grouped` (`GroupKFold`) and `temporal` (a fixed pre/post-2024 cutoff) are deterministic single partitions (`Scripts/rk_pipeline/splits.py`), so we compute no seed sweep for them. Model `random_state` (RF/GBM) is held fixed at 0, so this table isolates variance from which videos land in train vs. test across 5 stratified partitions, not full training variance (`outputs/experiments/phase1_multiseed.json`, `Scripts/phase1_multiseed.py`).

Two patterns stand out: (1) GBM is the strongest model on relevance, dominant tactic, and platform, and logistic regression on tool and family; (2) `family` and `tool` are weak under every combination tried (family never exceeds 0.13 macro-F1; tool peaks at 0.20 on the stratified split and reaches only 0.08 on the other two). Section 9.1's channel-identity check makes literal channel naming an unlikely explanation for the family result, leaving the weak-label ceiling (§4.3) and class sparsity (18-class family and 7-class tool taxonomies, several classes with single-digit support) as the remaining candidates, neither of which this paper isolates.

Table 3 reports only the single best (representation, model) cell per task. The full sweep has 144 trained combinations: TF-IDF and embeddings for all 5 tasks, plus structured

features for family and dominant tactic, each with 4 trained models and 3 CV protocols (`outputs/experiments/phase1_*.csv`). Figure 1 summarizes it, leaving out the circular dominant-tactic structured cells and taking the best remaining representation for each (task, model, split) cell so the figure stays legible, and Table 4 gives exact numbers for the stratified-random split.

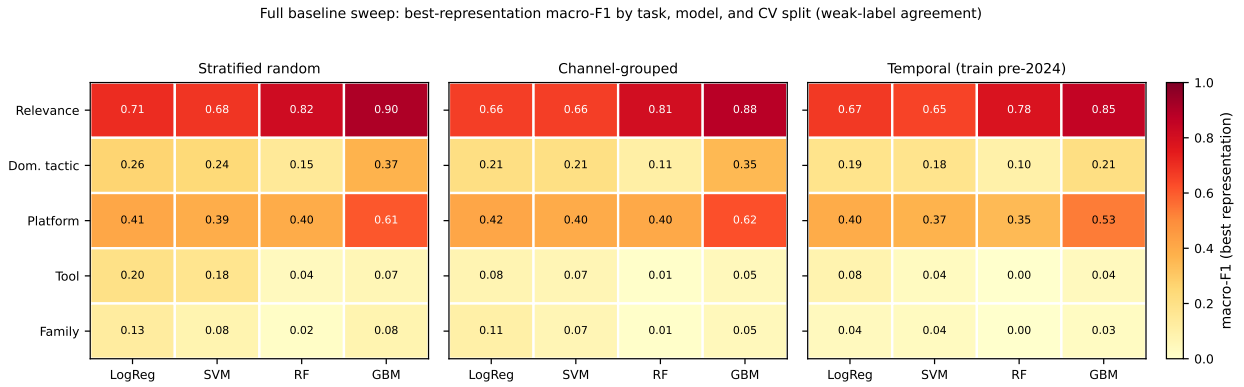


Figure 1: Best-representation macro-F1 for every (task, model) pair, one panel per CV protocol (weak-label agreement, §4.3). `family/tool` stay weak under every model and every split; `relevance` is both the strongest task and the most split-robust.

Table 4: Best-representation macro-F1 per (task, model), stratified-random split (weak-label agreement, §4.3), excluding dominant-tactic structured features. The temporal split costs some tasks far more than others, per Figure 1: the best family score falls from 0.13 to 0.04, tool from 0.20 to 0.08, and dominant tactic from 0.37 to 0.21, while platform (0.53) and relevance (0.85) hold up better.

Task	LogReg	SVM	RF	GBM
Relevance	0.71	0.68	0.82	0.90
Dominant tactic	0.26	0.25	0.15	0.37
Platform	0.41	0.39	0.40	0.61
Tool	0.20	0.18	0.04	0.07
Family	0.13	0.08	0.02	0.08

Figure 2 plots Table 4 as a 3D stem plot. Thin marked stems keep every value visible from any viewing angle, which a solid-bar rendering could not guarantee.

6. Uncertainty Quantification

Scope note (§4.3 applies here): calibration is normally reported against ground truth: does an 80%-confidence prediction turn out correct roughly 80% of the time? Without independent labels, the only available target is the weak-label distribution, which changes what a well-calibrated result means. A model perfectly calibrated against its own training signal has learned to echo a regex confidently, which does not make it a well-calibrated model of ransomware content. We report these results because reliability diagrams and Expected

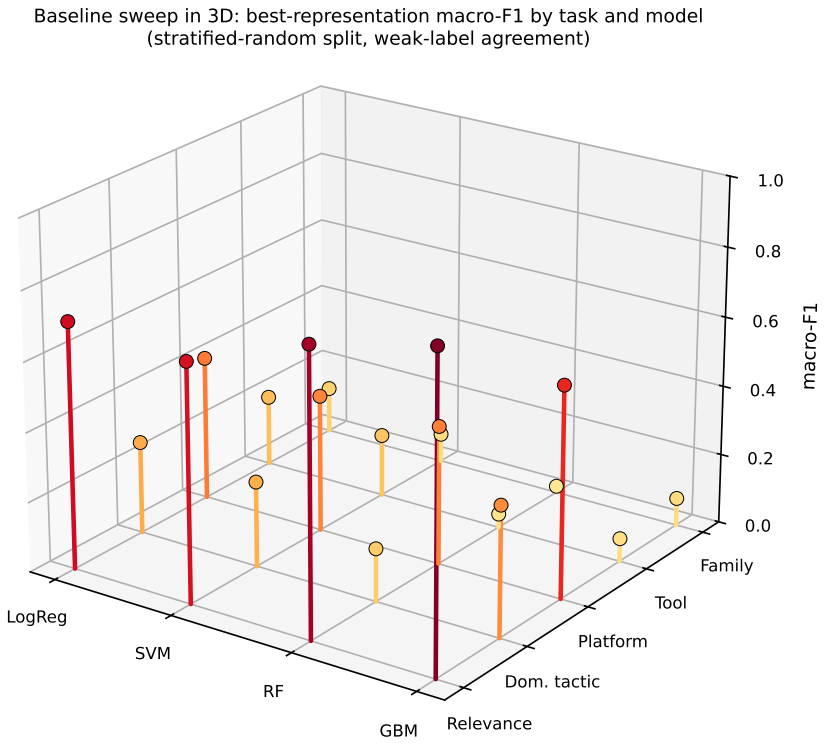


Figure 2: Table 4 as a 3D stem plot, with every point visible regardless of viewing angle.

Calibration Error against the weak labels still answer a narrower question: does a model's confidence track its *agreement with the extraction pipeline*, or is it uniformly overconfident regardless of agreement?

6.1. Calibration against weak labels

For a multiclass task, let $\hat{p}(\cdot \mid x)$ be the trained model's predicted class distribution for input x, and let $\hat{c}(x) = \arg\max_c \hat{p}(c \mid x)$ with confidence $\hat{p}_{\max}(x) = \max_c \hat{p}(c \mid x)$. Agreement with the weak label $y^{\mathrm{KA}}(x)$ (the Knowledge Agent's own extraction) is $a(x) = \mathbb{1}[\hat{c}(x) = y^{\mathrm{KA}}(x)]$. For the two multi-label tasks (`family`, `tool`), whose models are independent per-class binary classifiers (Section 5), each (sample, class) pair (x, k) is treated as its own calibration point, with confidence $\hat{p}_{\max}(x, k) = \max\big(\hat{p}_k(x), 1 - \hat{p}_k(x)\big)$ and agreement against the binary weak label thresholded at 0.5.

Reliability is summarized with B=10 equal-width confidence bins $\{I_b\}_{b=1}^{B}$ and the standard Expected Calibration Error,

$$\mathrm{ECE} \;=\; \sum_{b=1}^{B} \frac{|S_b|}{N} \left|\mathrm{acc}(S_b) - \mathrm{conf}(S_b)\right|, \qquad S_b = \{x : \hat{p}_{\max}(x) \in I_b\}, \tag{1}$$

where $\mathrm{acc}(S_b) = \frac{1}{|S_b|}\sum_{x \in S_b} a(x)$ and $\mathrm{conf}(S_b) = \frac{1}{|S_b|}\sum_{x \in S_b} \hat{p}_{\max}(x)$. Per the scope note above, $a(x)$ measures agreement with y^{KA}, not correctness, so Eq. (1) is a *weak-label ECE*. We also report the confident-disagreement rate at threshold $\tau = 0.9$,

$$\mathrm{CDR}_\tau \;=\; \frac{1}{N} \sum_{i=1}^{N} \mathbb{1}[\hat{p}_{\max}(x_i) > \tau] \cdot \big(1 - a(x_i)\big), \tag{2}$$

the closest available analogue to a "confidently wrong" rate without independently verified labels.

Each task is evaluated on the 5-fold stratified out-of-fold predictions of its best (feature, model) combination from Table 3.

Table 5: Weak-label calibration (Equations (1) and (2)) and the RF tree-vote entropy proxy (Equation (3)), best (feature, model) per task. N is the number of (sample) or (sample, class) calibration points. Entropy columns give mean entropy when the forest agrees and disagrees with the weak label, and how well entropy alone separates the two (AUROC, Algorithm 1).

Task	N	ECE	$\mathrm{CDR}_{0.9}$	Entropy (agr./dis.)	AUROC
Relevance	1034	0.079	0.071	0.910 / 0.980	0.763
Dominant tactic	1034	0.178	0.113	0.654 / 0.710	0.729
Platform	1034	0.110	0.104	0.716 / 0.770	0.682
Tool (per-label)	7238	0.016	0.016	0.121 / 0.275	0.977
Family (per-label)	18612	0.018	0.019	0.095 / –	–

Figure 3 plots the reliability curves underlying Table 5. Dominant tactic has the largest ECE (0.178): without the tactic counts, its model is markedly overconfident against the weak labels. Two cautions apply. First, per-bin counts are small (single digits to low tens

per bin at N=1034), so individual bin heights are noisy and the curves show direction only. Second, the low per-label ECE of `tool` and `family` reflects class imbalance more than skill. Most of their (sample, class) pairs are negative, so predicting "absent" agrees with the weak label almost everywhere, despite the low macro-F1 in Table 3, which weights classes equally rather than by prevalence. For the same reason `family`'s RF entropy proxy, which runs on TF-IDF rather than the task's best embedding representation, has no "disagree" entries to average.

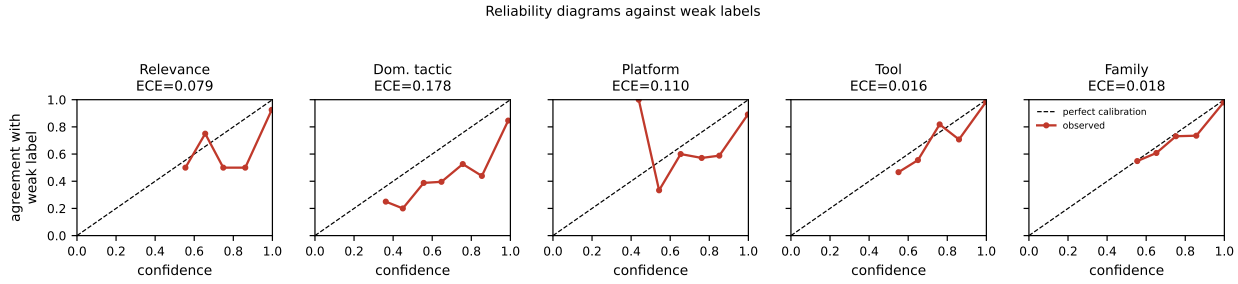


Figure 3: Reliability diagrams against weak labels, one panel per task with a usable calibration curve (Table 5). Diagonal is perfect calibration.

6.2. A free uncertainty proxy: Random Forest tree-vote entropy

Reliability diagrams need a held-out calibration protocol; a Random Forest offers a per-prediction uncertainty signal with no separate calibration step. For an input x and a forest of T trees producing individual (hard) predictions $t_1(x), \ldots, t_T(x)$ over K classes, let $\pi_c(x) = \frac{1}{T}\sum_{j=1}^{T} \mathbb{1}[t_j(x) = c]$ be the empirical vote share for class c. The normalized tree-vote entropy is

$$H(x) \;=\; -\frac{1}{\log K}\sum_{c=1}^{K} \pi_c(x)\,\log \pi_c(x) \;\in\; [0,1], \tag{3}$$

with $H(x)$=0 when every tree agrees and $H(x)$=1 at a uniform split across all K classes. For the multi-label tasks, K=2 per class and $H(x,k)$ is averaged over classes k for a per-sample summary. Algorithm 1 is the exact out-of-fold procedure used to produce Table 5's entropy columns.

The comment in Algorithm 1 records an implementation pitfall. scikit-learn [25] fits each tree in a `RandomForestClassifier` on internally label-encoded integer targets, so `tree.predict()` returns class *indices*, not the original label strings the ensemble's own `.predict()` returns. Comparing tree-level votes directly against string labels without mapping back through the fitted forest's class index silently produces 0% apparent agreement on every multiclass task. Our initial calibration run showed exactly that pattern across all three multiclass tasks, which is how we identified the bug. Table 5's entropy AUROC column shows the proxy works once fixed: 0.68–0.98 across tasks, best on `tool` (many easy true negatives) and weakest on `platform`.

Algorithm 1 RF tree-vote entropy, 5-fold out-of-fold

Require: dataset $\mathcal{D}$, task with label function $y(\cdot)$, feature builder ϕ

$\mathcal{F} \leftarrow$ 5 stratified folds of $\mathcal{D}$
for fold $(\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{test}}) \in \mathcal{F}$ **do**
    fit ϕ on $\mathcal{D}_{\text{train}}$; fit Random Forest $\{t_j\}_{j=1}^{T}$ on $\phi(\mathcal{D}_{\text{train}})$
    for $x \in \mathcal{D}_{\text{test}}$ **do**
        $t_j(x) \leftarrow$ individual tree prediction, $j = 1, \ldots, T$ ▷ sklearn returns internal integer-encoded classes here, not the original label strings; see note below
        recover original class labels via the fitted forest's class index before comparing to $y(x)$
        compute $H(x)$ via Equation (3); record $a(x) = \mathbb{1}[\text{forest majority vote} = y(x)]$
    end for
end for
return $\{(H(x), a(x))\}_{x \in \mathcal{D}}$, pooled across folds

7. Conformal Prediction

Scope note: split conformal prediction's distribution-free coverage guarantee concerns exchangeability with the calibration set, not label correctness. It holds when calibrated against weak labels because the guarantee is that the *weak* label falls in the predicted set $(1 - \alpha)$ of the time, not that the *real-world* label does. Conformal coverage against the Knowledge Agent's labels therefore validates agreement with the extraction pipeline, not correctness.

7.1. Method: split conformal with the LAC score

Split conformal prediction (14, 15; see 16 for a tutorial treatment) uses a nonconformity score $s(x, y)$ that is small when y is a plausible label for x. This paper uses the least-ambiguous-set (LAC) score,

$$s(x, y) \;=\; 1 - \hat{p}(y \mid x), \tag{4}$$

calibrated on a held-out split $\mathcal{D}_{\text{calib}}$ of size n disjoint from both the training and test data. The calibration threshold uses the standard finite-sample correction,

$$\hat{q} \;=\; \text{Quantile}\left(\{s(x_i, y_i)\}_{i=1}^{n} ;\; \frac{\lceil (n+1)(1-\alpha) \rceil}{n} \right), \tag{5}$$

and the prediction set for a new input x is

$$C(x) \;=\; \left\{ y \;:\; \hat{p}(y \mid x) \geq 1 - \hat{q} \right\}. \tag{6}$$

Under exchangeability of $\mathcal{D}_{\text{calib}}$ and the test point, this construction guarantees

$$\mathbb{P}[\, y_{\text{test}} \in C(x_{\text{test}}) \,] \;\geq\; 1 - \alpha, \tag{7}$$

with α=0.10 (90% target coverage) throughout. As the scope note above states, y_{test} here is the weak label y^{KA}: Equation (7) is a distribution-free guarantee about agreement with the

extraction pipeline, not about real-world correctness. For `dominant_tactic` and `platform`, Equations (4) to (6) apply directly over the task's full class vocabulary. `family` and `tool` are independent per-class binary classifiers (Section 5) with no single multiclass structure, so each class k gets its own binary LAC calibration and its own $\hat{q}_k$, and we report a sample's per-label prediction sets both individually and jointly (all labels simultaneously covered).

We run Algorithm 2 for 5 repeated random 60/20/20 train/calibration/test splits per task and average across repeats, because a single 20% calibration set at this sample size (n≈200) does not estimate $\hat{q}$ reliably.

Algorithm 2 Split conformal calibration and coverage evaluation

Require: dataset $\mathcal{D}$, task, target coverage $1-\alpha$, repeats R

for $r = 1, \ldots, R$ **do**
  split $\mathcal{D}$ into $\mathcal{D}_{\text{train}}$ (60%), $\mathcal{D}_{\text{calib}}$ (20%), $\mathcal{D}_{\text{test}}$ (20%), stratified
  fit model on $\mathcal{D}_{\text{train}}$
  compute $\hat{q}$ on $\mathcal{D}_{\text{calib}}$ via Equations (4) and (5)
  for $x \in \mathcal{D}_{\text{test}}$ **do**
    build $C(x)$ via Equation (6); record $\mathbb{1}[y^{\text{KA}}(x) \in C(x)]$ and $|C(x)|$
  end for
end for
pool test-set coverage indicators across all R repeats
report marginal coverage, and coverage restricted to the Year≥2026 and Conference-channel slices

7.2. Results

Table 6: Split-conformal coverage at 90% target (α=0.10), pooled over 5 repeats, n≈1035 pooled test predictions per task. `family`/`tool` report mean per-label coverage and the stricter all-labels-jointly-covered rate. Set size is out of the task's class count (K) for `dominant_tactic`/`platform`, or out of 2 (per label) for `family`/`tool`. `family` uses TF-IDF with logistic regression here rather than its best embedding representation.

Task	K	Coverage	Set size	Yr≥26	Conf.
Dominant tactic	11	0.908	2.60/11	0.913	0.926
Platform	4	0.925	1.19/4	0.932	0.899
Family (per-label)	18	0.911 (joint: 0.446)	0.93/2	0.487^{joint}	0.572^{joint}
Tool (per-label)	7	0.907 (joint: 0.650)	0.92/2	0.640^{joint}	0.752^{joint}

Figure 4 plots Table 6. First, does the guarantee hold empirically? Marginal coverage for `dominant_tactic` and `platform` (0.908, 0.925) meets the 0.90 target. Platform sets average 1.19 of 4 classes, while dominant-tactic sets average 2.60 of 11, which is the cost of the target coverage for a model this weak. Second, the conditional-coverage stress test: all four multiclass slice numbers stay within about 3 points of the marginal rate. These slices are subsets of the same random splits, so they also appear in the calibration data; they test coverage on corpus subgroups, not under distribution shift. `family` and `tool`'s per-label

coverage (0.91, 0.91) is close to target, but their *joint* (all-labels-simultaneously) coverage is far lower (0.45, 0.65). That is an expected consequence of accumulating misses across 18 and 7 binary calibrations, not a failure of Equation (7), which was never a joint guarantee. Both figures exceed what independent labels would give (about 0.19 and 0.51), so per-label misses are correlated. Joint coverage also varies more across slices than the multiclass rates do: on Conference channels it is 13 points above the marginal for `family` and 10 for `tool`. Per Section 7.1, all of the above describes agreement with y^{KA}.

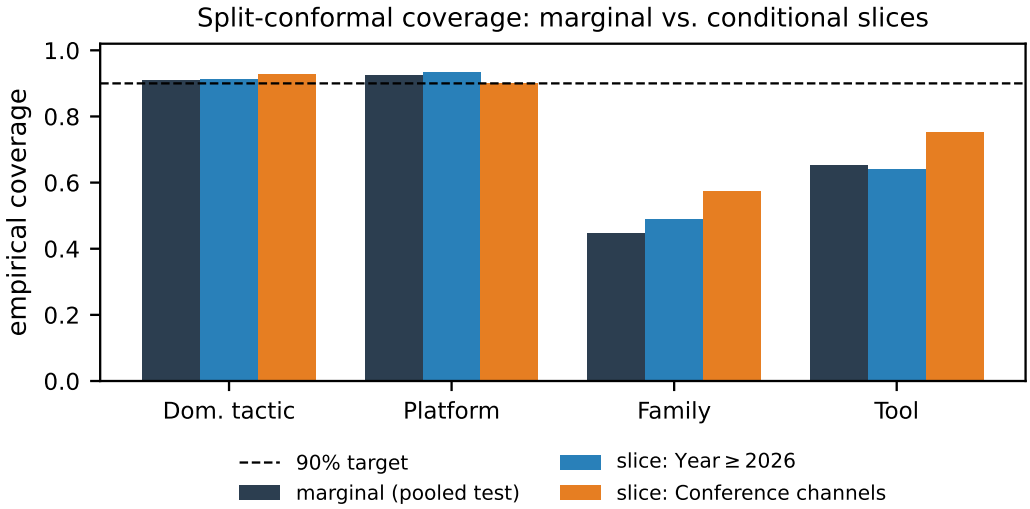


Figure 4: Empirical conformal coverage, marginal vs. the two conditional-coverage stress-test slices, against the 90% target (Table 6).

8. Out-of-Distribution Detection

OOD detection asks whether a model's own confidence or embedding-space behavior differs between in-distribution and out-of-distribution input. Scoring that needs no correct dominant-tactic label, since the question is which population an input came from. The weak labels are still involved: the shared classifier is trained on them, and the temporal and leave-rare-family setups define their populations with the regex's family list. Only the channel-type and external-corpus partitions are free of it.

8.1. Setups and methods

We use four setups, each with a stated in-distribution (ID)/out-of-distribution (OOD) split. Three come from the corpus's own structure rather than synthetic corruption, and the fourth is an external sanity check:

1. **Temporal emergence.** ID: pre-2024 videos. OOD: videos from 2024 or later mentioning a family whose first appearance in this corpus is 2024 or later. Algorithm 3's assertion step checks each family's first-occurrence year (Maze, Medusa, Black Basta, INC: 2024; Cactus: 2025). This is a family's first appearance in this corpus, not its real-world emergence.

2. **Channel-type shift.** ID: Vendor + DFIR channels. OOD: Conference + Independent channels.
3. **Leave-rare-family-out.** ID: videos never mentioning "Cl0p" (11 occurrences in total, unlike Conti, LockBit, Qilin, and Akira, each with 19 or more). OOD: the 11 Cl0p-mentioning videos.
4. **External corpus.** ID: RansKnow transcripts. OOD: 20 Newsgroups [26] (`rec.sport.baseball` + `sci.space`, excluding 38 documents left empty after header, footer, and quote removal), included as a sanity check that the methods below can detect an obviously unrelated domain before trusting them on setups 1–3.

Four detection methods use a shared `dominant_tactic` classifier (TF-IDF features with LightGBM, the best dominant-tactic result in Table 3 and a natural multiclass fit for the softmax-family methods below). We refit it on each setup's ID-only training pool so that no OOD signal leaks into fitting:

$$s_{\text{MSP}}(x) = 1 - \max_c \hat{p}(c \mid x) \tag{8}$$

$$s_{\text{energy}}(x) = -\log \sum_c \exp\big(f_c(x)\big) \tag{9}$$

$$s_{\text{maha}}(x) = \min_c \big(\phi(x) - \mu_c\big)^\top \Sigma^{-1} \big(\phi(x) - \mu_c\big) \tag{10}$$

$$s_{\text{conf}}(x) = 1 - \hat{p}\big(\hat{c}(x) \mid x\big) \qquad \text{(Equation (4) at the predicted class)} \tag{11}$$

Equation (8) is Hendrycks and Gimpel's max-softmax-probability baseline [17]. Equation (9) is the energy score [18], computed from LightGBM's raw pre-softmax margins $f_c(x)$ (*not* from $\hat{p}(c \mid x)$: probabilities already sum to 1, which makes Equation (9) degenerate to a constant if computed from them). Equation (10) is the class-conditional Gaussian Mahalanobis distance of Lee et al. [19], with $\phi(x)$ the 384-dim mean-pooled sentence embedding (Section 5's footnote) and μ_c, Σ the ID-train class-conditional means and pooled covariance, using the classifier's own *predicted* classes to form groups (consistent with Section 4.3: no independently verified labels exist to group by instead). Σ is a Ledoit-Wolf shrinkage estimate, because the temporal setup has only 162 training videos in 384 dimensions and its sample covariance is singular. Equation (11) reuses Section 7's LAC score, evaluated at the predicted rather than true class for OOD flagging. It is deliberately the same formula as Equation (8), but its classifier is refit on the 75% of the training pool left after holding out a calibration split, so its AUROCs differ slightly. We report it separately for comparability with Equation (5)'s calibrated threshold (does a point's own score exceed the 90%-target $\hat{q}$), which is the operationally distinct number.

The assertion in Algorithm 3 caught a drafting error. We first drafted the temporal setup's family list from memory as "Qilin, DragonForce, Black Basta," before checking first-occurrence years against the corpus. Qilin first occurs in 2019, not after 2023, and DragonForce does not appear in this corpus, so that list would have produced a meaningless setup. Table 7 uses the corrected list (Maze, Medusa, Black Basta, INC, Cactus), and Algorithm 3 verifies it on every run.

8.2. Results

Figure 5 plots Table 7. Three observations:

Algorithm 3 OOD setup evaluation

Require: corpus $\mathcal{D}$, ID/OOD partition rule for the setup

1: **assert** every claimed-emerging family's first-occurrence year matches the setup's stated cutoff ▷ caught a factual error at this step; see note below
2: split ID pool into $\mathcal{D}_{\text{train}}$ (75%) / $\mathcal{D}_{\text{test}}^{\text{ID}}$ (25%)
3: fit `dominant_tactic`/TF-IDF/LightGBM on $\mathcal{D}_{\text{train}}$
4: score $\mathcal{D}_{\text{test}}^{\text{ID}}$ and $\mathcal{D}_{\text{OOD}}$ under Equations (8) to (11)
5: report AUROC(ID vs. OOD) per method, plus the conformal exceedance-rate contrast

Table 7: OOD-detection AUROC by setup and method (Equations (8) to (11)). $n_{\text{ID}}/n_{\text{OOD}}$ are ID-test/OOD sample counts. Stratified bootstrap 95% intervals for every cell are in `outputs/phase4_ood.json`.

Setup	n_{ID}	n_{OOD}	MSP	Energy	Mahal.	Conf.
Temporal emergence	55	31	0.425	0.341	0.220	0.422
Channel-type shift	156	353	0.600	0.582	0.672	0.578
Leave-rare-family-out	256	11	0.304	0.196	0.145	0.238
External corpus (sanity)	259	1152	0.622	0.477	0.999	0.627

The confidence-based methods (MSP, energy, conformal) fail the sanity check. On `external_corpus`, MSP reaches only 0.622 (95% interval 0.573–0.672) and energy 0.477. The predictions show why: 99.0% of 20 Newsgroups documents are classified `None` (no dominant tactic detected) by the classifier, against 18.5% of the ID test set, and with *higher* average confidence (mean $\hat{p}_{\max}$: 0.922 OOD vs. 0.895 ID). A sparse, off-topic TF-IDF vector and a tactic-free ransomware transcript both produce the same signature, confidently "no tactics detected," so the classifier cannot distinguish "nothing here" from "wrong domain entirely." This is a known blind spot of softmax/probability-based OOD detection in general [17], reproduced here. These methods reach 0.58–0.60 on channel-type shift and fall below 0.5 on the other two setups.

Mahalanobis distance on sentence embeddings passes the sanity check but inverts on the family-based setups. It reaches 0.999 (95% interval 0.996–1.000) on `external_corpus` and 0.672 (0.618–0.723) on channel-type shift, where Conference/Independent channels differ detectably in embedding space; Section 8.3 shows that both figures are optimistic and gives more conservative values. On temporal emergence (0.220, 0.130–0.324) and leave-rare-family-out (0.145, 0.065–0.244) it is below chance with intervals that exclude 0.5, so these are not noise: OOD videos sit *closer* to the class means than ID videos do. The covariance estimate is not the cause, since the earlier pseudo-inverse estimate gives 0.223 and 0.145. We tested two explanations. The first is selection: every OOD video in these setups names a family, against 25% and 20% of their ID test sets. Restricting ID to family-naming videos (n=14, n=51) raises the scores to 0.353 (0.191–0.532) and 0.257 (0.119–0.417), so the temporal inversion is no longer distinguishable from chance, while the leave-rare-family inversion narrows but persists. The second is length: mean-pooling more chunks pulls long videos toward the class means (Spearman ρ between distance and word count is -0.66 and -0.71), and OOD videos are longer (median 6,288 vs. 2,950 words and 6,480 vs. 5,036). Adjusting log distance for log word count leaves 0.256

and 0.148, so length does not explain the inversion by itself. The combined effect of the two remains untested, and with n_{OOD}=11 the leave-rare-family interval stays wide. Because both populations are selected through the regex family list, these two setups are a weak test of whether a detector notices novel families.

The conformal exceedance-rate contrast (Equation (5)'s $\hat{q}$) is uninformative here by construction. The calibrated threshold is at least 0.997 in every setup, because the classifier assigns the weak label almost no probability on more than a tenth of its calibration points (n between 41 and 194). The OOD-flagging score is taken at the *predicted* class (Equation (11)), so it cannot exceed $1 - 1/K$, about 0.91 with 11 classes, and no point can cross the threshold. Equation (11)'s AUROC column is the more informative number from this method; the zero exceedance rates are in the released JSON (`outputs/phase4_ood.json`).

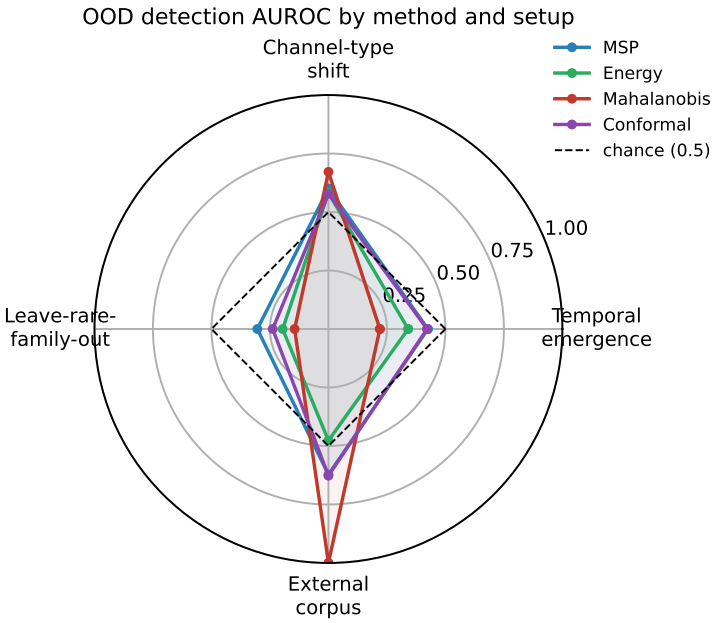


Figure 5: OOD-detection AUROC by method (color) and setup (axis), against the chance line at 0.5 (Table 7). Mahalanobis is the only method to clear chance by a wide margin, on the channel-type and external setups, and falls well below it on the other two.

Each AUROC summarizes a comparison between two score distributions, and the two figures below show more of that comparison. Figure 6 plots the MSP score density for the ID-test and OOD populations, per setup, as a 3D ridgeline: the blue (ID) and red (OOD) curves overlap heavily and both peak in the low-score, high-confidence region, the visual counterpart of MSP's weak AUROCs. Figure 7 is a 3D t-SNE projection of the same 384-dim sentence embeddings the Mahalanobis method (Equation (10)) computes its distances on, colored by ID/OOD membership. OOD points are scattered through the ID cloud in all three setups shown. t-SNE distorts distances, so this is illustrative only: it does not show the separation the Mahalanobis AUROCs measure on channel-type shift, and should not be read as evidence for or against it.

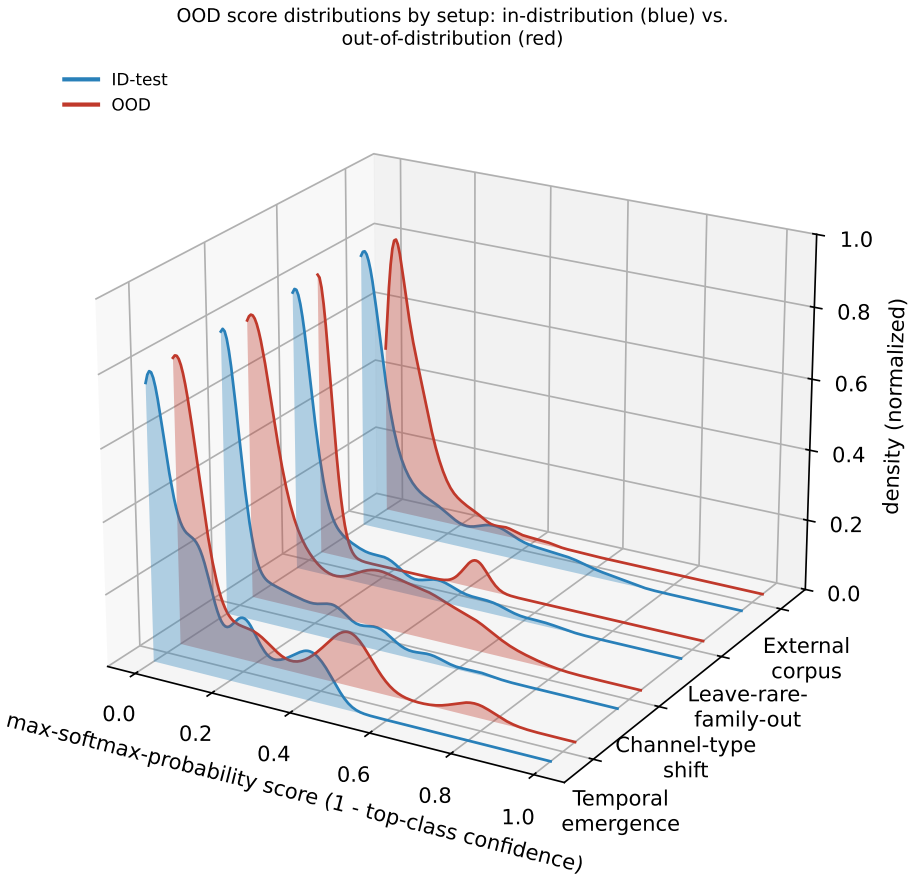


Figure 6: ID-test vs. OOD max-softmax-probability score density per setup, as a 3D ridgeline (kernel density estimate, not binned histograms). Overlap between the blue and red curves is the visual counterpart of MSP's weak AUROCs.

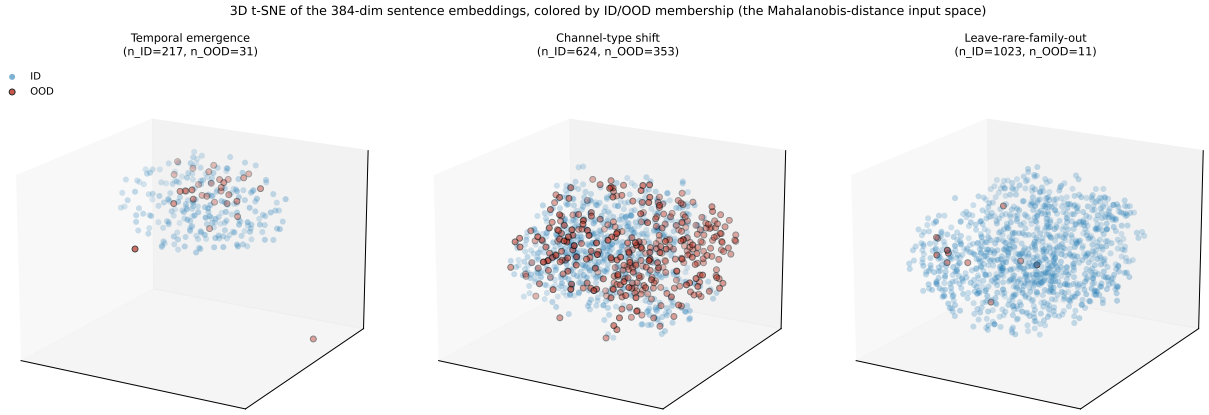


Figure 7: 3D t-SNE of the 384-dim sentence embeddings per setup, colored by ID/OOD membership, for the three within-corpus setups. Illustrative only, since t-SNE distorts distances: no setup shows visibly separated clusters.

8.3. Leakage and confound checks

We audited these results for information shared between training and evaluation (`Scripts/phase4_leakage_audit.py`). No video or URL appears in both training and evaluation in any setup, and every fitted component (vectorizer, classifier, class means, covariance) uses training data only. Four indirect effects were tested.

Shared channels. The ID split is random by video, so 98–100% of ID test videos share a channel with training, while the OOD channels of channel-type shift never do, which favors separation. Holding ID channels out lowers its mean Mahalanobis AUROC over 5 seeds from 0.677 to 0.644 (range 0.608–0.688), so the 0.672 in Table 7 is modestly inflated but the effect survives. The inversions also survive: temporal emergence stays at 0.199, or 0.204 when OOD is limited to channels unseen in training, and leave-rare-family-out at 0.124. Every Cl0p video comes from a channel that also appears in training, so that setup cannot be held out on both sides.

Duplicates. Eight transcript pairs exceed 0.98 cosine similarity, including the four videos listed twice (Section 3). Dropping evaluation videos with a near-duplicate in training changes no Mahalanobis AUROC by more than 0.004.

Unmatched family mentions. Four of the 217 pre-2024 ID videos contain a term matching an emerging family that the context-gated regex did not tag. Removing them moves temporal emergence's Mahalanobis AUROC from 0.220 to 0.261, and no ID video contains a Cl0p term.

Length and formatting in the sanity check. The stored transcripts contain timestamp lines and have a median of 4,985 words, while the 20 Newsgroups posts have no timestamps and a median of 74 words, and Mahalanobis distance falls as documents get longer. Removing timestamps from the corpus and re-embedding it moves the sanity check only from 0.999 to 0.996. Truncating each ID test video to its first 200-word chunk as well lowers it to 0.943 (0.930–0.955), which we take as the fairer figure. It is conservative, because the same truncation alone separates ID videos from their own full-length versions at AUROC 0.994. Timestamp removal moves the other Mahalanobis AUROCs by at most 0.022. Effects of timestamps on the TF-IDF baselines in Section 5 were not separately tested.

Taken together, Section 8.2 does not support a general claim that this pipeline's models reliably detect distributional shift. It supports a narrower one: softmax-confidence methods have an identifiable blind spot on this classifier (the "no signal" vs. "wrong domain" confusion above), embedding distance detects a change of domain and, more modestly, of channel type, and neither approach shows that it notices a video naming a family absent from training.

9. Interpretability and Perturbation Robustness

Unlike Sections 5–7, this section asks about model *behavior*: does it exploit shortcuts, what does it rely on, and how much do its predictions move under semantically irrelevant edits? These questions need no independent ground truth, so Section 4.3's scope note applies only where a model's training labels matter.

9.1. Confound checks

Length. Transcript word count correlates moderately with `Tactic_Total_Mentions` (Pearson $r = 0.409$) and weakly with `Family_Count` ($r = 0.131$), and not significantly with

`Tool_Total_Mentions` ($r = 0.043$), so the confound is not uniform across signal types. Dividing every tactic's count for one video by that video's word count cannot change which tactic has the highest count, because argmax is scale-invariant per row, so the confound cannot affect how the `Dominant_Tactic` label was derived. The remaining question is whether it inflates a model given those counts. Re-scoring logistic regression on structured features with length-normalized (mentions per 1,000 words) instead of raw counts gives 0.753 macro-F1 against 0.764, a 1.1-point difference, so the model is not exploiting length. This check covers only the structured representation, which Table 3 excludes for this task as circular; length is not tested for the TF-IDF model.

Channel identity. 486 of 1,034 transcripts (47%) contain at least one literal mention of their own channel's name. Masking those mentions and re-scoring TF-IDF LogReg under both CV protocols leaves the `channel_grouped`-vs-`stratified_random` gap unchanged for `family` (0.028 raw → 0.031 masked) and for `dominant_tactic` (0.020 raw → 0.029 masked). Whatever a model exploits when a channel is not held out, it is not the channel literally naming itself: the leakage this split protocol exists to catch is stylistic or topical, and name-masking does not address it.

9.2. Counterfactual explanations

Finding A raises a question: does a classifier trained on the *fixed* labels still show the regex's original word-collision fragility? We trained TF-IDF+LogReg models on the corrected `Family` labels in 5 stratified folds and scored every video only with a model that did not train on it, because the videos in question are labeled not-`Play` in the training data. The test set is the 275 videos the Section 4.1 fix removed from `Play`, meaning the fix's own judgment rather than a per-video human review. Of these, 8 (2.9%) still reach $P(\text{Play}) \geq 0.5$, with a maximum of 0.999. Their median probability is 0.005, against 0.001 for the 648 other not-`Play` videos, and the model ranks them above those videos with AUROC 0.58 (standard error about 0.02). That is a weak trace of the collision, neither a clean result nor wholesale inheritance. It should be read against a weak model: only 18.9% of the 111 videos labeled `Play` reach 0.5, with a median of 0.019.

We then asked what drives the model's confident `Play` predictions, using a greedy word-*type* removal counterfactual search (remove the vocabulary word whose removal most reduces $P(\text{Play})$, repeat until the prediction flips or a budget is exhausted) on the 8 labeled `Play` videos with the highest held-out probability (all at least 0.97). `play` appears in 3 of 8 removal sets, and 0 of 8 cases flip on removing "play" alone. Every case flips within 1–8 removals, mostly generic words rather than ransomware-specific vocabulary: the ASR filler "uh" appears in 5 of 8 sets, and one case flips on removing "uh" alone. The model's confident predictions therefore rest on a diffuse, easily disturbed set of words rather than one trigger word or a clear ransomware vocabulary, a finding about the decision boundary that is separate from the word-collision question.

9.3. Perturbation robustness

As an untargeted counterpart to the counterfactual search, we apply two perturbation types and measure flip rate (does the top prediction change) and mean absolute probability drift on freshly fit TF-IDF+LogReg models. *Simulated ASR noise* (compound proper nouns split apart, e.g. `WannaCry`→"wanna cry," `Mimikatz`→"mimi cats," plus 8% random token

deletion) is applied to the 74 official-caption videos, to simulate what those transcripts would look like after the ASR cascade that produced the other 92.7% of the corpus (Finding B). *Synonym substitution* (common non-jargon words swapped for synonyms) is applied to a 150-video sample.

`dominant_tactic`, `platform`, and `relevance` are essentially unmoved by either perturbation (0.0–0.7% flip rate, mean drift < 0.01 in every case). `family` flips 20–26% of the time under both perturbations despite a much smaller mean drift ($\sim$0.002) than any other task. This is the weak-model issue from Table 3 (`family`'s best macro-F1 is 0.129) appearing in another form: closely bunched, low-confidence probabilities mean a negligible nudge can change which class wins. We did not run a gradient-based adversarial attack, because no real adversary submits transcripts to evade this classifier in production. The practical risks are ASR transcription noise and channel-style variation, which this sweep and the channel-grouped protocol address. Together with Section 8's temporal-emergence and leave-rare-family-out setups, these results describe how the system handles evolving, novel, and noisy input.

10. Limitations and Ethics

The most consequential limitation of this paper is the absence of independent ground truth. The other limitations (sampling bias, ASR noise, source rights, dual use) are of a more familiar kind, but the ground-truth gap changes what every quantitative claim in Sections 5–7 *means*, not only how confidently it can be stated.

- **No independent ground truth.** This project has no human-annotated evaluation set independent of the Knowledge Agent pipeline (Section 4.3). Recruiting external annotators would have required ethics approval beyond the project's existing approval for public data, and was not undertaken; we considered a single-investigator annotation pass and rejected it as too close to the system under evaluation to count as independent evidence. Every result in Sections 5–7 is therefore agreement with an audited but imperfect extraction pipeline (Finding A), not verified accuracy. Section 8's OOD scores and Section 9's behavioral diagnostics are less directly affected, although two OOD setups and the `Play` check are defined using the regex's labels. Independent annotation, with whatever ethics review the authors' institution requires, is the highest-value extension of this work, since none of the modeling in Sections 5–7 can currently be checked against anything but itself.

- **Sampling bias.** The corpus is conditioned on which channels operators chose to run, which videos they chose to title and describe in keyword-matchable ways, and YouTube's own recommendation and search surfacing, so it is not a random or exhaustive sample of ransomware-relevant discourse.

- **ASR noise.** 92.7% of transcripts are machine-transcribed, mostly by `whisper_base`. Section 9.3's simulated-ASR-noise sweep bounds how much this moves predictions; Finding B's word-error-rate-on-jargon question (comparing ASR output against source audio directly, rather than a simulation of ASR-style errors) remains open.

- **Weak-label ceiling.** Every quantitative result in Sections 5–7 is bounded by the Knowledge Agent's own precision/recall, which Finding A shows is not perfect even after the disambiguation fix, and which Section 9.2 shows leaving a weak trace in at least one trained model.

- **Source rights and platform terms.** The transcripts derive from talks, webinars, and walkthroughs whose speakers and channel operators hold their own rights, and they were obtained through the YouTube Data API, `yt-dlp`, and audio download for Whisper transcription. The release contains transcript text and metadata but no audio or video. A video being public does not by itself grant a right to redistribute its speech as text, and we did not obtain permission from individual channels or speakers. YouTube's terms of service may also restrict automated collection of this kind, so re-users should check the terms that apply to them. The authors' CC0 dedication covers only the authors' own contributions (the extracted features, labels, and compiled metadata). It grants no re-use rights in the transcript text or the underlying talks, which remain with their owners and are shared here for research use. A rights holder can have a transcript removed from future releases by opening an issue on the repository or by contacting the corresponding author.

- **Dual use.** This corpus covers ransomware tactics, tools, and case studies from public educational content. Every underlying video is already public and indexed by YouTube's search, and was produced by defenders (vendors, DFIR practitioners, conference speakers) for other defenders. RansKnow discloses no technique, tool, or tactic that is not already public. The MITRE ATT&CK knowledge base and the CTI datasets in Section 2 [8, 9] already structure comparable information at finer granularity and higher label quality than RansKnow's weak-label extraction (Section 4.3). RansKnow adds a different modality (transcribed spoken narration rather than written reports) and a scale accessible without specialized CTI tooling. The realistic marginal risk is a small reduction in the effort needed to *aggregate* already-scattered public information, the same tradeoff every public threat-intelligence aggregation resource makes. Given Section 4.3's weak-label ceiling and Section 5's weak family and tool classification, the corpus currently favors defensive research uses (weak-supervision methodology, corpus-level trend analysis) far more than offensive use, for which primary-source technical documentation would be a more direct and reliable starting point.

11. Conclusion

We presented RansKnow, an open MITRE ATT&CK-aligned corpus of 1,128 ransomware-related YouTube video records, 1,034 with transcripts. The audit shows that a plausible regex label can be dominated by detections a simple context check rejects: 71% of pre-fix `Play` detections (Table 1). A corpus released without such an audit would carry this error silently.

Against the audited labels, models reproduce relevance well (macro-F1 0.90) but dominant tactic poorly (0.38). In out-of-distribution detection, softmax methods fail a sanity check that Mahalanobis distance passes, yet Mahalanobis distance scores below chance when out-of-distribution videos are selected for naming a family. The audit and the counterfactual

and perturbation checks can be reused on any dataset labeled by keyword or regex extraction over scraped or transcribed text.

The priority for future work is independent human annotation of a subset of the labels (Section 10). Until then, the baseline, calibration, and conformal results describe agreement with the pipeline: a macro-F1 of 0.90 on `relevance` means a LightGBM classifier on TF-IDF reproduces the regex output at that rate, not that it identifies relevant videos at that rate.

CRediT authorship contribution statement

Henry Kabuye: Conceptualization, Methodology, Software, Data curation, Formal analysis, Investigation, Writing – original draft. **Biju Issac:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare no competing interests.

Data availability

The RansKnow corpus, the extraction and audit pipeline, and all experiment code are publicly available at https://github.com/Henrinnes/RansKnow and on Kaggle at https://www.kaggle.com/datasets/henrykabuye/ransknow-v1. The transcripts derive from third-party videos; Section 10 describes the rights that remain with their owners.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used a generative AI coding assistant in order to write and debug analysis code, rerun and check experiments, and edit the manuscript text. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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Appendix A. Keyword List

The 21-term inclusion keyword list, matched case-insensitively against video title and description (Section 3), verbatim from `Scripts/fetch_transcripts.py`'s `KEYWORDS` constant:

```
ransomware, extortion, double extortion, encrypt, encryption, decrypt,
data leak, incident response, "ir", lockbit, alphv, blackcat, conti,
ryuk, cl0p, clop, akira, revil, sodinokibi, ransomware-as-a-service,
raas
```

The bare token "`ir`" is matched with surrounding spaces (`" ir "`) specifically to avoid matching it as a substring of unrelated words.